

```

ubuntu@ip-172-31-21-110: ~
ubuntu@ip-172-31-21-110: ~ 81x24

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-21-110:~$ netstat -ltn
Active Internet connections (only servers)

```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	127.0.0.53:53	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:53	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:53	0.0.0.0:*	LISTEN
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:953	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.1:953	0.0.0.0:*	LISTEN
tcp	0	0	127.0.0.54:53	0.0.0.0:*	LISTEN
tcp	0	0	172.31.21.110:53	0.0.0.0:*	LISTEN
tcp	0	0	172.31.21.110:53	0.0.0.0:*	LISTEN
tcp6	0	0	fe80::8ff:fcff:fedc::53	:::*	LISTEN
tcp6	0	0	fe80::8ff:fcff:fedc::53	:::*	LISTEN
tcp6	0	0	:::1:53	:::*	LISTEN
tcp6	0	0	:::1:53	:::*	LISTEN
tcp6	0	0	:::22	:::*	LISTEN
tcp6	0	0	:::1:953	:::*	LISTEN
tcp6	0	0	:::1:953	:::*	LISTEN

```

ubuntu@ip-172-31-21-110:~$

```

```
ubuntu@ip-172-31-21-110: ~
ubuntu@ip-172-31-21-110: ~ 81x24
ubuntu@ip-172-31-21-110:~$ sudo apt install net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 77 not upgraded.
Need to get 204 kB of archives.
After this operation, 811 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 net
-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Fetched 204 kB in 0s (12.4 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 71833 files and directories currently installed.)
Preparing to unpack ../net-tools_2.10-0.1ubuntu4.4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

```
ubuntu@ip-172-31-21-110: ~
ubuntu@ip-172-31-21-110: ~ 81x24
bind9: unrecognized service
ubuntu@ip-172-31-21-110:~$ sudo service bind9 status
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: en
   Active: active (running) since Thu 2026-01-08 11:09:27 UTC; 34s ago
     Docs: man:named(8)
   Main PID: 2055 (named)
    Status: "running"
     Tasks: 8 (limit: 1008)
    Memory: 23.0M (peak: 23.4M)
       CPU: 52ms
    CGroup: /system.slice/named.service
            └─2055 /usr/sbin/named -f -u bind

Jan 08 11:09:27 ip-172-31-21-110 named[2055]: network unreachable resolving './>
Jan 08 11:09:27 ip-172-31-21-110 named[2055]: network unreachable resolving './>
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Jan 08 11:09:27 ip-172-31-21-110 named[2055]: network unreachable resolving './>
Jan 08 11:09:27 ip-172-31-21-110 named[2055]: managed-keys-zone: Initializing a
Jan 08 11:09:27 ip-172-31-21-110 named[2055]: managed-keys-zone: Initializing a
ubuntu@ip-172-31-21-110:~$ sudo apt install net-tools
```

```
ubuntu@ip-172-31-21-110: ~  
ubuntu@ip-172-31-21-110: ~ 81x24  
  
ubuntu@ip-172-31-21-110:~$ sudo apt install bind9  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  bind9-utils dns-root-data  
Suggested packages:  
  bind-doc  
The following NEW packages will be installed:  
  bind9 bind9-utils dns-root-data  
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.  
Need to get 423 kB of archives.  
After this operation, 1631 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Err:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 bin  
d9-utils amd64 1:9.18.39-0ubuntu0.24.04.1  
  404 Not Found [IP: 54.144.148.213 80]  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dns  
-root-data all 2024071801~ubuntu0.24.04.1 [5918 B]  
Err:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 bin  
d9 amd64 1:9.18.39-0ubuntu0.24.04.1  
  404 Not Found [IP: 54.144.148.213 80]  
Fetched 5918 B in 0s (119 kB/s)  
E: Failed to fetch http://us-east-1.ec2.archive.ubuntu.com/ubuntu/pool/main/b/bin
```

EC2

Grupos de seguridad

sg-0653da8b6a1e5177f - launch-wizard-10

Panel

Vista global de EC2

Eventos

Instancias

Tipos de instancia

Plantillas de lanzamiento

Solicitudes de spot

Savings Plans

Instancias reservadas

Alojamientos dedicados

Reservas de capacidad

Capacity Manager

Imágenes

AMI

Catálogo de AMI

Elastic Block Store

Volúmenes

Instantáneas

Administrador del ciclo de vida

Red y seguridad

Security Groups

Direcciones IP elásticas

Grupos de ubicación

Pares de claves

Interfaces de red

Las reglas del grupo de seguridad de entrada se han modificado correctamente en el grupo de seguridad (sg-0653da8b6a1e5177f) [launch-wizard-10]

Detalles

Nombre del grupo de seguridad

launch-wizard-10

ID del grupo de seguridad

sg-0653da8b6a1e5177f

Descripción

launch-wizard-10 created 2026-01-08T09:56:49.716Z

ID de la VPC

vpc-02ba8f7225454a3d5e

Propietario

991377533768

Número de reglas de entrada

3 Entradas de permisos

Número de reglas de salida

1 Entrada de permiso

Reglas de entrada

Reglas de salida

Compartiendo

Asociaciones de VPC

Etiquetas

Reglas de entrada (3)

Administrar etiquetas

Editar reglas de entrada

	Name	ID de la regla del gr...	Versión de IP	Tipo	Protocolo	Intervalo de puertos	Origen	Descripción
☐	-	sgr-0e626cfa4e8bb397	IPv4	TCP personalizado	TCP	953	0.0.0.0/0	-
☐	-	sgr-09e819a095b309262	IPv4	DNS (TCP)	TCP	53	0.0.0.0/0	dns 53
☐	-	sgr-0f40a02bf15ebc9aa	IPv4	SSH	TCP	22	0.0.0.0/0	dns 53

/etc/bind/

- Configura el servidor Bind9 como un servidor caché DNS apuntando a las DNS de Google (8.8.8.8).

```
ubuntu@ip-172-31-21-110: /etc/bind
ubuntu@ip-172-31-21-110: /etc/bind 81x24

options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    forwarders {
        8.8.8.8;
    };

    //=====

    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //=====

    dnssec-validation auto;
}

"named.conf.options" 24L, 837B 14,10-24 Top
```

- Configura una instancia nueva como cliente en AWS, denominada **DNS-cliente** y configúralo para que las DNS las resuelva DNS-servidor.

Resumen de instancia de i-04feba8a1891169ac (clienteDNS) Información

Se ha actualizado hace less than a minute

ID de la instancia i-04feba8a1891169ac	Dirección IPv4 pública 54.242.243.107 dirección abierta	Direcciones IPv4 privadas 172.31.18.16
Dirección IPv6 -	Estado de la instancia En ejecución	DNS público ec2-54-242-243-107.compute-1.amazonaws.com dirección abierta
Tipo de nombre de anfitrión Nombre de IP: ip-172-31-18-16.ec2.internal	Nombre DNS de IP privada (solo IPv4) ip-172-31-18-16.ec2.internal	Direcciones IP elásticas -
Responder al nombre DNS de recurso privado IPv4 (A)	Tipo de instancia t3.micro	Hallazgo de AWS Compute Optimizer Suscribirse a AWS Compute Optimizer para recibir recomendaciones. Más información
Dirección IP asignada automáticamente 54.242.243.107 [IP pública]	ID de VPC vpc-026a8f225454a3d3e	Nombre del grupo de Auto Scaling -
Rol de IAM -	ID de subred subnet-0474c45bec1ba3c34	Administradas falso
IMDSv2 Required	ARN de instancia arn:aws:ec2:us-east-1:891377333768:instance/i-04feba8a1891169ac	
Operador -		

Detalles Estado y alarmas Monitoreo Seguridad Redes Almacenamiento Etiquetas

▼ Detalles de la instancia Información

ID de AMI ami-0ecb62995f68bb549	Monitoreo desactivado	Detalles de la plataforma Linux/UNIX
Nombre de AMI ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20251022	Imagen permitida -	Protección de terminación desactivado
Detener la protección desactivado	Hora de lanzamiento Thu Jan 08 2026 12:29:28 GMT+0100 (hora estándar de Europa central)	Ubicación de AMI amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-serve

4. Realiza consultas desde tu cliente entrando en la web de 3 dominios distintos, y utilizando los comandos **nslookup**, **dig**, **ping**, etc...

```
ubuntu@ip-172-31-18-16: ~  
ubuntu@ip-172-31-18-16: ~ 82x24  
^Cubuntu@ip-172-31-18-16:~$ dig @52.201.56.85 www.iespoligonosur.org  
; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> @52.201.56.85 www.iespoligonosur.org  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: REFUSED, id: 63950  
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1  
;; WARNING: recursion requested but not available  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: 30e417ce394b962b01000000695f989194a37d9c6e468280 (good)  
; EDE: 18 (Prohibited)  
;; QUESTION SECTION:  
;www.iespoligonosur.org.                IN      A  
  
;; Query time: 0 msec  
;; SERVER: 52.201.56.85#53(52.201.56.85) (UDP)  
;; WHEN: Thu Jan 08 11:44:17 UTC 2026  
;; MSG SIZE rcvd: 85  
  
ubuntu@ip-172-31-18-16:~$
```

```
ubuntu@ip-172-31-18-16: ~  
ubuntu@ip-172-31-18-16: ~ 83x26  
ubuntu@ip-172-31-18-16:~$ dig @172.31.21.110 www.iespoligonosur.org  
; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> @172.31.21.110 www.iespoligonosur.org  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 61206  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: 6f6a450263e5720301000000695f997752050bf1677c1c6e (good)  
;; QUESTION SECTION:  
;www.iespoligonosur.org.                IN      A  
;; ANSWER SECTION:  
www.iespoligonosur.org. 2905     IN      A      212.227.230.62  
;; Query time: 7 msec  
;; SERVER: 172.31.21.110#53(172.31.21.110) (UDP)  
;; WHEN: Thu Jan 08 11:48:07 UTC 2026  
;; MSG SIZE rcvd: 95  
ubuntu@ip-172-31-18-16:~$
```